CpE-371: SOFTWARE ENGINEERING AND DEVELOPMENT

UniReserve

UniReserve

Software Requirements Specification (SRS)

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Prepared by	Noura Almethen (Team Leader), 2221154924 Mouddy Almowsharji, 2221162836 Shaikha Almandeel, 2221164096

Department of Computer Engineering
College of Engineering and Petroleum
Kuwait University
2025 - 2026

Signature Page

I did my share of the work, and I have a general understanding of the contents of the assignment

Name	Signature
----------Noura Almethen----------	------Noura------
----------Mouddy Almousharji----------	-----Mouddy-----
----------Shaikha Almandeal----------	-----Shaikha-----

Report Evaluation

Criterion	Evidence / focus	Weight	Marks	Comments
Document control & professionalism	Version history, ownership, formatting, completeness	5		
Introduction & project framing	Purpose, scope, vision, constraints, assumptions, stakeholders	10		
Functional requirements	Clear, grouped, numbered, testable 'shall' statements	15		
Use-case model	Diagram, actors, use-case specs, traceability, coverage	15		
Non-functional requirements	Measurable NFRs with verification method	10		
Domain / data model	Domain overview, ERD/model, retention/deletion/auditability	10		
UI prototype & usability	Scenario coverage, key screens, findings, updates	10		
Project management & evidence	WBS, Gantt, refs, repo links, meeting records	5		
Team Deliverable Quality (Total)				**out of 80**

Individual Grade (20%)

Student name	Avg peer score (/28)	Adjustment	Final mark (/100)	Notes

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List of Abbreviations

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1. Document Control

1.1 Version history

Version	Date	Author	Change summary	Approval
0.1	2026-03-28	Noura Almethen	Initial draft of Sections 1, 2, 3	Pending
0.2	2026-03-29	Team	Added Introduction, requirements, use case model, and formatting updates	Pending
1.0	2026-03-30	Noura Almethen	Final reviewed version for submission	Approved

1.2 Credits

Name	Role/Task
Noura Almethen	Use Case Diagram, and textual specifications for the 3 main use cases
Mouddy Almowsharji	Functional Requirements and Non-Functional Requirements
Shaikha Almandeel	Section 2 (Purpose, Product Vision, Scope, Assumptions and Dependencies, Constraints, Stakeholders, Overview)

1.3 Related documents / links

Backlog / Trello board: UniReserve_R1-Part1_Trello_Board

2. Introduction

2.1 Purpose

This Software Requirement specification document describes the functional and non-functional requirements of the UniReserve system. The purpose of this document is to make sure everyone involved in the project, like developers, team members, instructors, and stakeholders are on the same page when it comes to how the system works and what it needs to do. It's a guide for designing, building, and testing the system, so we can make sure it does what it's supposed to do. by having this document, we can all agree on what the system should be able to do, and that helps us avoid confusion later on.

2.2 Product vision

UniReserve is a booking system designed to help students, staff, and faculty members reserve rooms and equipment on campus. The system helps solve problems of informal booking methods, such as sending messages or making manual requests, which can lead to issues like conflicts and double bookings. UniReserve allows users to search for available rooms, send requests, and receive approvals. This makes the booking process more efficient and reduces errors and mistakes that happen when finding rooms, equipment, and makes it better and easier for users. By using UniReserve, users can have a smoother experience on campus without facing many problems.

2.3 Scope

In scope:

- Search real-time availability of rooms and equipment
- Submit booking requests with details
- Admin approval /rejection system
- Prevent double booking
- Modify or cancel bookings
- Maintain booking history

Out of scope:

- Payment system
- Mobile application (for now)
- Full production deployment
- Integration with external systems

2.4 Assumptions and dependencies

Assumption:

- Users have valid university accounts
- Internet access is available
- Users understand basic system usage

Dependency:

- Database system for storing bookings
- University authentication system
- Server availability for system operation.

2.5 Constraints

[Describe the relevant constraints]

- [Regulatory / policy constraints]
- [Technology/platform constraints]
- [Operational constraints (deployment environment)]

2.6 Stakeholder summary

Role	**Primary goals**	**Key needs / pain points**	**Priority (H/M/L)**
Student	Book rooms/equipment	Easy booking, no conflicts	H
Staff faculty	Manage bookings	Reliable approval system	H
Admin	Control system	Avoid double bookings, manage rules	H

2.7 Overview

This document is divided into several sections. Section 2 introduces the system's purpose, scope, and stakeholders. Section 3 presents the functional requirements and uses case model of the system. Each requirement is clearly defined to support system development and testing. The document provides a complete understanding of the UniReserve system.

3. Functional Requirements and Use-case Model

The UniReserve system is designed to support a structured and transparent workflow for managing room and equipment reservations. Each requirement is labeled FR-01 through FR-07 to ensure clarity and traceability. Together, they define the essential functions that the system must provide.

- FR-01: Availability Search

The system shall enable users to search for rooms and equipment by specifying the date, start time, end time, and required capacity. It shall clearly display whether each resource is available or already booked, and provide detailed information such as type, location, and current booking status.

• FR-02: Reservation Request Management

The system shall allow students, staff, and faculty to submit reservation requests. Each request must include all necessary details such as resource type, date, time, and purpose. A unique identifier shall be assigned to every request, which will initially be stored with a Pending status until reviewed by the Facilities Admin.

• FR-03: Policy and Validation Rules

The system shall enforce booking policies to prevent misuse and ensure fairness. It shall block double-bookings, apply permission rules for restricted resources, and enforce time-related constraints such as maximum booking duration, advance booking limits, and cancellation deadlines. Requests that violate these rules shall be rejected with a clear explanation provided to the user.

• FR-04: Admin Review and Decision

The system shall allow the facilities Admin to view all pending requests, examine their details, and make decisions. The admin shall be able to approve valid requests or reject invalid ones with a documented reason. Once a decision is made, the system shall update the reservation status accordingly.

• FR-05: Booking Confirmation and Audit Trail

The system shall notify requesters of the outcome of their requests. For approved reservations, a confirmation record shall be created. All actions, including submission, approval, rejection, cancellation, and modification, shall be logged in an audit trail to ensure accountability and transparency.

• FR-06: Cancellation and Modification

The system shall allow users to cancel or modify their bookings within the permitted deadlines. Any modification shall trigger a revalidation process to check for conflicts and compliance with policies. Attempts to cancel or modify bookings after the deadline shall be blocked.

• FR-07: Notifications and Reminder Schedule

The system shall send notifications at key stages: when a request is submitted, when a decision is made, and when a booking is cancelled or modified. For approved reservations, reminders shall be scheduled 24 hours and 1 hour before the booking start time. If contact details are missing or invalid, the system shall mark the notification as failed.

3.2 Use Case Model

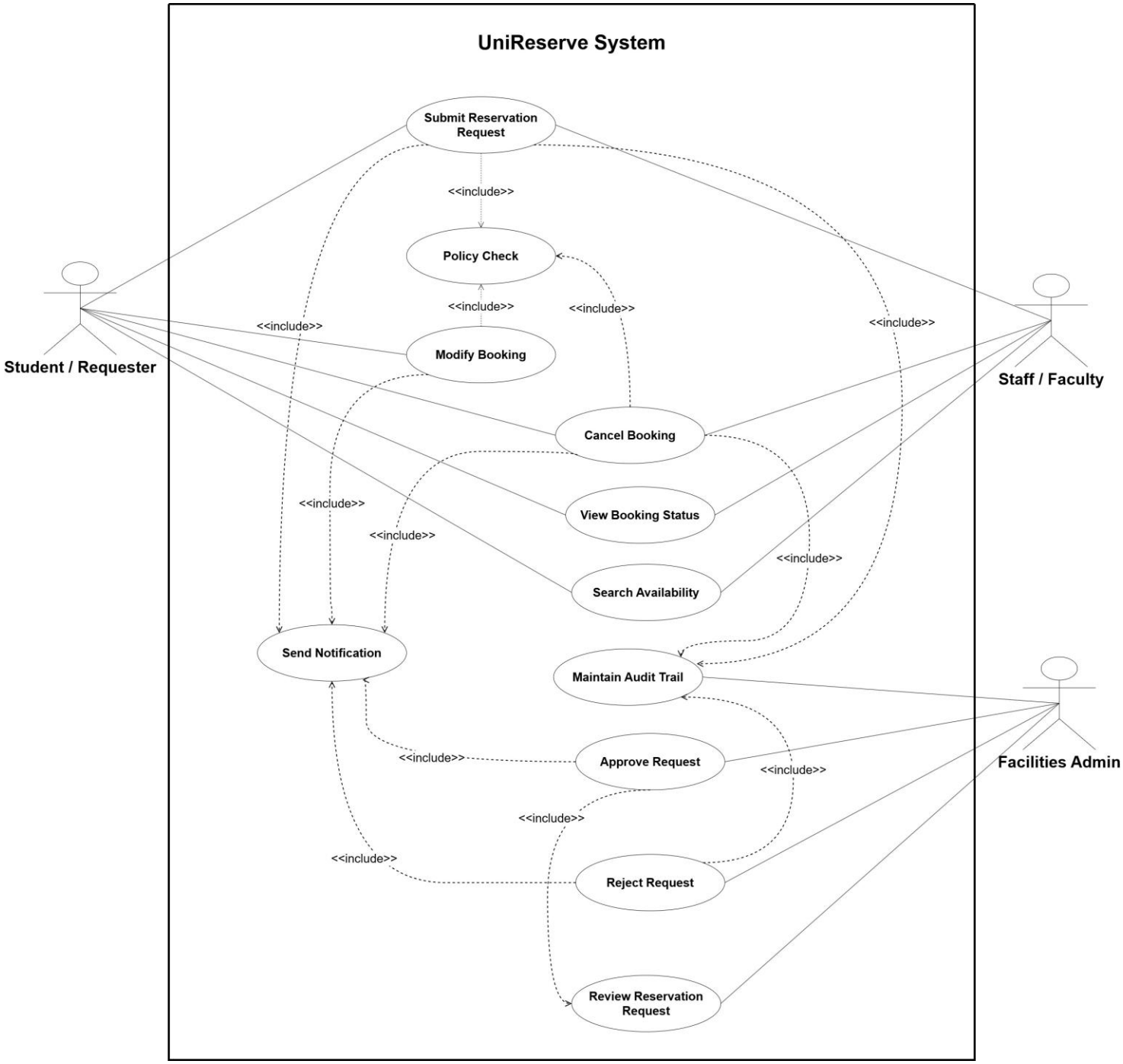

Figure 1 Use Case Diagram

Use case UC- 01**:** Search Availability

Related Requirements	*FR-01*
Description (Goal)	*Allow the user to search for available rooms or equipment for a selected date and time.*
Primary Actor(s)	*Student, Staff/Faculty*
Secondary Actor(s)	*None*
Preconditions	• *The user can access the system.* • *Rooms and equipment are already registered in the system.*
Post-conditions	• *A list of matching resources is displayed.* • *The user can see whether each resource is available or already booked.*
Related use cases	*UC-02: Submit Reservation Request*
Basic Flow	1. *The user opens the availability search page.* 2. *The user enters the date, start time, end time, and required capacity or resource type.* 3. *The system checks room and equipment availability.* 4. *The system displays matching resources with their availability status and details.* 5. *The user reviews the search results.*
Alternative Flow	• *A1: No matching resources are found. The system displays a message that no resources are available for the selected period.* • *A2: The user enters an invalid time range. The system asks the user to correct the input.*

Use case UC- 02**:** Submit Reservation Request

Related Requirements	*FR-02, FR-03, FR-07*
Description (Goal)	*Allow the user to submit a reservation request for a selected room or equipment.*
Primary Actor(s)	*Student, Staff/Faculty*
Secondary Actor(s)	*Facilities Admin*
Preconditions	• *The user has selected a resource.* • *The requested booking date and time have been entered.*
Post-conditions	• *A reservation request is stored with a unique identifier and Pending status, or the request is rejected if it violates booking policies.* • *A submission notification is generated.*
Related use cases	*UC-01: Search Availability* *UC-03: Review and Approve/Reject Request*
Basic Flow	1. *The user selects an available room or equipment.* 2. *The user enters the booking details, including date, time, and purpose.* 3. *The user submits the reservation request.* 4. *The system checks booking policies and validates the request.* 5. *The system assigns a unique identifier to the request.* 6. *The system stores the request with Pending status.* 7. *The system sends a notification confirming that the request was submitted.*
Alternative Flow	• *A1: The selected resource becomes unavailable before submission. The system rejects the request and informs the user.* • *A2: The request violates a booking policy. The system rejects the request and displays the reason.* • *A3: Required fields are missing. The system asks the user to complete the missing information.*

Use case UC- 03**:** Review and Approve/Reject Request

Related Requirements	*FR-04, FR-05, FR-07*
Description (Goal)	*Allow the Facilities Admin to review pending reservation requests and approve or reject them.*
Primary Actor(s)	*Facilities Admin*
Secondary Actor(s)	*Student, Staff/Faculty*
Preconditions	• *A pending reservation request exists in the system.*
Post-conditions	• *The request status becomes Approved or Rejected.* • *The action is recorded in the audit trail.* • *A decision notification is sent to the requester.*
Related use cases	*UC-02: Submit Reservation Request*
Basic Flow	1. *The Facilities Admin opens the list of pending requests.The user enters the booking details, including date, time, and purpose.* 2. *The Facilities Admin selects one request.The system checks booking policies and validates the request.* 3. *The system displays the request details.* 4. *The Facilities Admin reviews the request.* 5. *The Facilities Admin approves the request.* 6. *The system updates the status to Approved.* 7. *The system creates a confirmation record.* 8. *The system logs the action in the audit trail.* 9. *The system sends an approval notification to the requester.*
Alternative Flow	• *A1: The Facilities Admin rejects the request. The admin enters a rejection reason, the system changes the status to Rejected, logs the action, and sends a rejection notification.* • *A2: A conflict or rule violation is detected during review. The system blocks approval and informs the admin.* • *A3: The Facilities Admin exits without making a decision. The request remains Pending.*

4. Non-functional Requirements

4. Non-Functional Requirements

In addition to the Functional Requirements that need to be satisfied by the UniReserve system, there are a set of Non-Functional Requirements (NFRs) that need to be met. Each NFR is presented along with a measurable criterion and verification method.

- NFR-01: Performance

1.1 Response Time The system shall respond to search queries within 3 seconds on average, assuming regular usage.

Verification: Performance testing.

- NFR-02: Reliability

2.1 System Availability: The system must provide 99% availability during normal business hours.

Verification: Monitoring and log analysis.

- NFR-03: Privacy and Ethics

3.1 Data Privacy: The software program will store user information securely to prevent any possible misuse.

Verification: Security inspection and penetration testing.

3.2 Ethical Use The use of booking data shall be restricted to within the eReservation system for academic and administrative purposes only.

Verification: Policy review and compliance audit.

- NFR-04: Usability

4.1 Interface Simplicity (Reduction) The interface shall enable users to complete a booking request in less than five steps.

Verification: Usability testing with representative users.

- NFR-05: Security

5.1 For all users, the system must require valid login credentials.

Verification: Security testing.

5.2: All sensitive information (such as user ids and booking details) will be encrypted.

Verification: Code inspection and encryption analysis.

5.3 Access Control: Access to administrative functions should be restricted to authorized personnel only.

Verification: Role-based access testing.

- NFR-06: Maintainability

6.1 Code Modularity (RD&M 6.1)

The system shall be developed with modular components to facilitate updates and enhancements.

Verification: Code review and inspection.

6.2 Error Logging The system shall record and store all error messages with sufficient detail to allow for diagnosis and maintenance of the system.

Verification: Monitoring and log inspection.

5. Domain overview

[Describe the core domain concepts and how they relate. Keep this aligned with the entity model.]

5.2 Entity-relationship diagram (ERD) / domain model

[Insert ERD/domain diagram here.]

5.3 Data retention, deletion, and auditability

- Retention: [e.g., keep audit logs for X days]
- Deletion: [e.g., GDPR delete within X days]
- Audit: [what events must be logged]

6. UI Prototype and Usability Evaluation

If the product is UI-heavy, include a prototype and a brief evaluation to reduce ambiguity and rework.

[Create a UI prototype covering a scenario for each main use case (e.g. if you have 3 main use cases, cover 3 scenarios, and so on) using Invision app. Create the screens using some proper software tool then connect them on Invision app. Make sure to show data in the scenarios (e.g. in case of a login form show it first empty then when the user clicks on it show the data filled in the fields).]

The goal of this prototype is elicit requirement from the customer using prototyping based on scenarios from the requirement specifications. This prototype is be able to handle *< your number of main use cases >* scenarios. It was tested on [3] users.

6.1 Prototype link(s)

- Figma / prototype: [link]

6.2 Key screens and states

[Create a prototype of your system on invision app covering the < your number of main use cases > *chosen scenarios. Create the screens using proper software tools then connect them on Invision app. Make sure to show data in the scenarios. Include the invision app link here. In addition, describe your preliminary design by presenting pictures of important screens and briefly explaining the function of each screen.]*

This section of the document contains brief descriptions of the main screens in *<name of your application>.* Each screen in the system shows the behavior and state changes that the user will experience when pressing a button or choosing an option on the screen. InVision App online prototyping tool was used to create all of the system screens. The UI interactive prototype is available online at this *<your invision app link>*.

- **…… Screen (e.g. Login Screen)**

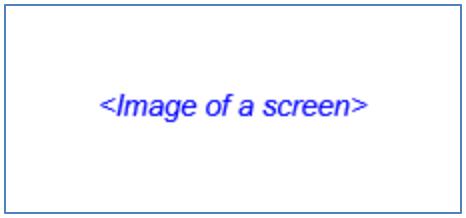

Figure 2 .... Screen

Description: ……..

- **…… Screen**

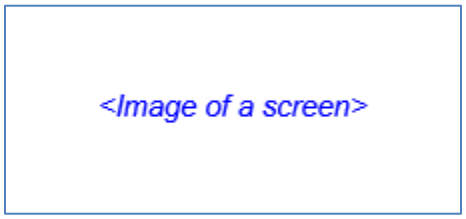

Figure 3 ..... Screen

Description: ……..

6.3 Usability evaluation

Scenario ID	Scenario	Participants	Findings	Action / requirement update
UE-0X	[User tries to …]	[N users, profile]	[What worked / what failed]	[Change to story/NFR/UI]
UE-0X	[User tries to …]	[N users, profile]	[What worked / what failed]	[Change to story/NFR/UI]
UE-0X	[User tries to …]	[N users, profile]	[What worked / what failed]	[Change to story/NFR/UI]
UE-0X	[User tries to …]	[N users, profile]	[What worked / what failed]	[Change to story/NFR/UI]

7. Appendix

7.1 References

- Reference 1: [standard / policy / paper / link]
- Reference 2: [link]

7.2 Project Management

Work Breakdown Structure

[Identify the activities, milestones and deliverables (like this document), responsibilities, duration, and dependencies. Be sure to provide sufficient detail in the structure and identify clear deliverables (a table that shows the tasks numbers, names, duration, start date, finish date, predecessors, and resources). Include all the phases of the project starting from the proposal and finishing by the final submission of the project.]

Gantt Chart

[The Gantt Chart should show the start and finish dates of subtasks and summary tasks of your project. It should also show the duration of each task and the resources (team members) assigned to each task.]

Add repo link

Team Meetings

Meeting Minutes (R1-Part 1): UniReserve_Meeting_Minutes_R1_Part1